

FLATTRIKE · REQUIREMENTS

FlatTrike requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

FlatTrike requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see FTK-PRB-001). The status column gives the TRL 2 concept's position against each target from the first-order estimates in FTK-PRC-001. Four requirements are **not met** by the current concept: R8 (empty mass), R10 (stability with no cargo), R13 (hill climb without assist) and R15 when the electric assist option is included.

The **design load case** used throughout is an 80 kg rider and 150 kg of cargo in the box on a trike of about 68 kg: about 300 kg gross, on a flat dry road unless stated.

Table 1. FlatTrike requirements and concept status.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R1	Carry the rated load	150 kg of cargo in the box plus a rider up to 90 kg; gross mass 300 kg or less, within the EN 17860 cargo cycle class	Mass budget; load case review	Met, no margin: about 298 kg with an 80 kg rider
R2	Frame strength	Frame survives a static proof load of twice the rated payload with no permanent set, and a fatigue load case representative of commercial use under EN 17860	Hand calculation and FEA of plates and bolted joints; later proof test	Unverified; first-order plate stresses are low (about 26 MPa), fatigue at bolt holes is the open risk
R3	No welding	Every steel part is flat 3 mm plate cut from a DXF file by laser or waterjet; no bending, welding or brazing; joints bolted with locknuts	Design review of the model and part list	Met by design
R4	One standard sheet	All steel frame parts nest on one 1,250 x 2,500 x 3 mm sheet (4 x 8 ft class)	Nesting study	Likely met: about 1.5 m ² of parts on a 3.1 m ² sheet
R5	Standard bicycle parts	Wheels (20 in, ISO 406), hubs, brakes, headset (1 1/8 in), bottom bracket, cranks, chain (1/8 in), seatpost (27.2 mm) and saddle are standard sizes sold in regional markets	Parts availability survey with the partner	Met by design, except the head tube and bottom bracket housings (see FTK-PRC-001)
R6	Ships flat	Complete kit, wheels included, packs into 0.4 m ³ or less with no piece longer than 1.0 m	Packing study	Likely met: about 1.0 x 0.8 x 0.45 m, about 0.36 m ³
R7	Assembled with hand tools	Two people assemble the kit in 4 h or less with spanners to 17 mm, hex keys, cone spanners and a screwdriver; no jig, because the plates locate each other with tabs and slots	Assembly sequence review; later timed build	Unverified
R8	Light enough to ride and push	Empty mass 55 kg or less, matching the improved rickshaw documented by ITDP	Mass estimate; later weighing	Not met: about 68 kg (plates about 30 kg, plywood box about 16 kg)
R9	Fits market lanes	Overall width 1.0 m or less; length 2.2 m or less; turning circle 6 m or less	Model check	Met: about 0.96 m wide, 2.11 m long, turning circle about 5.8 m
R10	Resist tipping in turns	Lateral acceleration at the tipping point 0.30 g or more, both loaded and with the rider only	Stability calculation; later tilt-table test	Not met with no cargo: about 0.18 g; met loaded, about 0.38 g
R11	Stop and park safely	Stop from 15 km/h in 6 m or less at gross mass on a dry road; drum brakes on all three wheels; a parking brake that holds the loaded trike on a 10 % grade	Braking calculation; later field test	Unverified; the required deceleration (about 1.5 m/s ²) is modest, but drum fade on long descents is a risk
R12	Pedal on the flat	Cruise at 7 km/h or more at gross mass on a flat dirt road with 110 W at the pedals	Power calculation	Met: about 7.6 km/h

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status (estimate)
R13	Climb local hills	Climb a 5 % grade for 100 m at 4 km/h or more at gross mass without the rider dismounting	Power and gearing calculation	Not met without assist: needs about 210 W at the wheel, about twice sustained rider output; met with the optional 250 W assist
R14	Repairable locally	Any single plate or standard part replaced in 30 min or less with hand tools; open DXF files let any laser shop re-cut one plate	Design review; later repair trial	Met by design; unverified in practice
R15	Affordable prototype	Prototype parts cost \$800 or less (<code>project.yaml</code> budget)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Met for the pedal-only base, about \$640. Not met with the optional electric assist, about \$1,040
R16	Last in commercial service	Frame life of 5 years or more in daily outdoor commercial use, against 2 to 3 years for traditional wooden rickshaw frames (ITDP)	Corrosion and fatigue review; later field trial	Unverified
R17	Work as a stall	Lockable box of 150 L or more; box lid usable as a counter at 0.75 to 0.95 m height	Model check	Met: about 190 L inside; lid at about 0.79 m

■ Assumptions

- Rolling resistance coefficient about 0.015 on a mix of worn asphalt and packed dirt; drag area about 0.9 m² for an upright rider behind a cargo box; air density 1.2 kg/m³.
- The rider sustains about 110 W at the pedals for an hour; short efforts can reach 200 W or more.
- Drivetrain efficiency (chain and 3-speed hub) about 89 %.
- Plate is 3 mm hot-rolled mild steel (S235JR or ASTM A36 class), density 7,850 kg/m³.
- Plywood is 9 to 12 mm exterior grade, about 700 kg/m³.
- The 300 kg gross limit follows the EN 17860 class for single and multi-track cargo cycles; the standard's detailed test loads have not yet been checked against this design.